Martin **Farkas**

PhD Candidate · Computer Engineering · Critical Systems Research Group, BME

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About

Martin Farkas is a PhD student in the Critical Systems Research Group at the Budapest University of Technology and Economics, where he completed his MSc with honors in 2025. His research is on the **model-driven design of decentralized trust solutions**: applying formal modeling and verification techniques to digital identity systems built on Self-Sovereign Identity standards and Zero-Knowledge Proofs. The aim is to make trust assumptions in such systems explicit and machine-checkable, rather than implicit and audited after the fact.

Research Interests

Formal verification of decentralized trust systems · Self-Sovereign Identity (SSI) and Verifiable Credentials · model-driven engineering for digital identity · business process modeling (BPMN) and security analysis · Zero-Knowledge Proofs and privacy-preserving cryptographic protocols.

Education

PhD in Computer Engineering
Budapest, Hungary

Budapest University of Technology and Economics
2025 — Present

- Topic: *Model-driven design of decentralized trust solutions*
- Group: Critical Systems Research Group
- Department: Artificial Intelligence and Systems Engineering (MIT)
- Supervisor: Dr. Imre Kocsis

MSc in Computer Engineering
Budapest, Hungary

Budapest University of Technology and Economics
2023 — 2025

- Specialization: IT Security, Critical Systems
- Final grade: **Summa Cum Laude**
- Thesis: *Self-Sovereign Identity based Self-Evaluated Policies*

BSc in Computer Engineering
Budapest, Hungary

Budapest University of Technology and Economics
2019 — 2023

- Specialization: System Engineering
- Thesis: *Self-Sovereign Identity supported payment on the openCBDC platform*

Secondary School Diploma
Győr, Hungary

Révai Miklós Secondary Grammar School
2013 — 2019

Academic Positions

Research Assistant
Budapest, Hungary

Critical Systems Research Group, BME
2023 — Present

- Privacy-preserving protocols for blockchain and SSI systems
- Zero-Knowledge Proof systems for verifiable credentials
- Course delivery in Self-Sovereign Identity, distributed systems, and blockchain
- Mentoring BSc and MSc students on SSI and blockchain research

Publications

A Self-Orchestration Model for Business Collaborations with Verifiable Process History Credentials — Seville, Spain
Farkas M., Péter B.Z., Kocsis I. · BPM 2025 Forum · LNBIP 564, Springer · doi:10.1007/978-3-032-02929-4_4 — August 2025

A Prolog-based Approach to Self-Evaluated, Declarative and Zero-Knowledge Verifiable Policies — Krakow, Poland
Farkas M., Toldi B.Á., Péter B.Z., Kocsis I. · EuroCyberSec Workshop · MASCOTS 2024, IEEE — October 2024

Design Space Exploration of Verifiable Credential Schemas using Partial Graph Modeling — Szeged, Hungary
Farkas M., Péter B.Z., Kocsis I. · CS² · 14th Conference of PhD Students in Computer Science · Volume of Short Papers — July 2024

Service

Sub-reviewer · ICSE 2026 (Demonstrations track) — for Grace Lewis · Nov 2025

Sub-reviewer · SAC 2026 — for Imre Kocsis · Nov 2025

Sub-reviewer · SafeComp 2026 — for Zoltán Micskei · Apr 2026

Awards & Honours

3rd place — National Scientific Students' Conference (OTDK) — Cybersecurity section
37th OTDK · *Self-evaluated policies using Zero-Knowledge Proofs* — April 2025

3rd place — BME VIK Students' Scientific Conference (TDK) — System Modelling section
Design Space Exploration of Verifiable Credential Schemas using Partial Graph Modeling — November 2024

1st place — BME VIK Students' Scientific Conference (TDK) — Information Systems section
Self-evaluated policies using Zero-Knowledge proofs — November 2023

Special prize — BME VIK Students' Scientific Conference (TDK) — Information Systems section
Payments in openCBDC with Self-Sovereign Identities — from the verifiable to the private — November 2022

Summa Cum Laude
MSc in Computer Engineering, BME — 2025

Research Projects

MNB–BME CBDC Cooperation — Budapest, Hungary
Hungarian National Bank & BME — 2024 — Present
Integrating an SSI-based Zero-Knowledge Proof protocol into a Central Bank Digital Currency prototype, extending earlier work in the Critical Systems Research Group.

EMAP — Elektronikus Munkaadat Szolgáltatási Platform
Budapest, Hungary

Nemzeti Adó- és Vámhivatal (NAV) · ET 3.1 — Sep — Nov 2025

Self-Sovereign Identity authorization system using Hyperledger Fabric

Applied SSI research project — Mar — Oct 2025

DC4EU — Digital Credentials for Europe
BME

Pilot facilitation — 2025

DigitalTech EDIH — Blockchain task at BME

European Digital Innovation Hubs initiative · European Commission · coursework development on permissioned blockchains and digital identity (SSI) — 2023 — Present

Training SMEs for the Digital Decade (SME4DD)

EU SME4DD project · coursework development for the digital identity management module — 2023 — Present

Industry Experience

Lead Engineer & Crypto Engineer
Budapest, Hungary

QUANOPT Kft. · Internet of People (IoP) SSI industry project — Summer 2025

- Lead engineer and crypto engineer from the QUANOPT side on porting the IoP Stack's Morpheus framework (gatekeeper-free DID and Verifiable Credential management) from the Hydra dPoS blockchain onto Hyperledger Fabric
- Reimplemented Morpheus's DID state machine and credential operations as Fabric chaincode, replacing Hydra's Layer-2 block-event model with Fabric's endorsement flow

Systems Engineer
Budapest, Hungary

QUANOPT Kft. — 2023

- Lead developer on a Hyperledger Fabric chaincode for metering-data storage at a Hungarian gas provider (Fablo, Java); built the Swagger REST API on top (Node.js, TypeScript) and ran the performance measurement campaign

Cryptocurrency Mining Platform Developer
Budapest, Hungary

NAS Kft. — 2022

- Prototyped a GPU-based cryptocurrency mining platform on Go Ethereum
- Conducted candidate mining-software review and market analysis informing platform design choices

Teaching & Supervision

BSc thesis & TDK supervision (BME VIK)

Cloud-ready, compositionally verifiable zero-knowledge evaluation of declarative policies — 2025

TDK supervision (BME VIK)

LLM-assisted creation and refinement of error propagation analysis models — 2025

Student mentoring

BSc and MSc students · SSI, distributed systems, blockchain — 2023 — Present

Supervising student research projects on Zero-Knowledge Proof learning paths and DeFi/blockchain topics.

Professional Development

BSidesBUD 2026 Budapest, Hungary
10th anniversary edition · community-driven IT security conference, Central Europe April 2026

Marktoberdorf Summer School (MOD25) Marktoberdorf, Germany
Specification and verification of secure cyberspaces August 2025

Hacktivity 2025 Budapest, Hungary
Central & Eastern Europe's longest-running IT security festival 2025

HUSTEF 2024 Budapest, Hungary
Hungarian Software Testing Forum 2024

ITBN Conf-Expo 2024 Budapest, Hungary
value chAIned · Informatikai Biztonság Napja · Hungary's IT security conference September 2024

Technical Skills

Formal Methods	**Tamarin Prover**, **Refinery**, epistemic logic, model-driven engineering, SysML, UML, PlantUML
Cryptography	**Zero-Knowledge Proofs**, Circom, Halo2, snark.js, zkVMs, Hyperledger AnonCreds, W3C Verifiable Credentials, DIDComm
Blockchain & DLT	**Hyperledger Fabric**, Hyperledger Aries, Hyperledger Indy, Go Ethereum, Solidity, Truffle, Ganache
Programming	**Rust**, **TypeScript**, **Python**, Java, Kotlin, C, C++, C#, JavaScript, Solidity, Prolog, Circom, Elixir, Bash
Web & Backend	Node.js, NestJS, Express, Next.js, Spring, Tailwind CSS, REST/Swagger
DevOps & Data	Docker, Kubernetes, Git, GitHub Actions, Jenkins, SQL, MongoDB
Languages	**Hungarian (Native)**, **English (C1)**